

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
4	(a)			Alpha presents <u>no</u> danger or it can't pass through the skin (1) Beta and gamma can get into the body (1) Beta is more <u>ionising</u> [than gamma so more dangerous] (1) Treat statements about cancer as being neutral	3			3		
	(b)	(i)		Finding the ratio of LLW to reactors in UK = $\frac{30\,100}{15} = 2007$ (1) For France, ratio = $\frac{87\,000}{58} = 1\,500$ so not valid (1) [Conclusion must be present to award 2 marks] Alternative: Ratio of reactors France to UK is $\frac{58}{15} = 3.87$ (1) Ratio of LLW France to UK = $\frac{87\,000}{30\,100} = 2.89$ so not valid (1) [Conclusion must be present to award 2 marks] Alternative: Ratio of reactors France to UK is $\frac{58}{15} = 3.87$ (1) Expected LLW in France: $30\,100 \times 3.87 = 116\,487$ so not valid (1) [Conclusion must be present to award 2 marks]			2	2	2	
		(ii)	I	The presence of volcanoes (1) Creates a danger of leakage from the containers (1)			2	2		
			II	Large population in a small area / population density (1) means there is less likelihood of remote areas being available as storage sites or being stored far away from people (1)			2	2		